lúbadh

_travelers guide
_for navigating time loops

2	10	4
decks · stereo link	min per loop, per deck	taps per deck

[rec] press **[rec]** hold **[shift]** = retrig/shift opp. = the other deck's button › then
◆ = preset / mode only

every control is per deck: left and right are identical

combos shown for one deck · mirror them for the other

amber = hold

01 transport — lbh_01

[rec]	record › press again to stop & play. press while playing = **overdub**
[erase]	erase this deck's loop
[shift]	jump back to start
turn **[speed]**	speed & direction: center = stall, cw = forward to 4×, ccw = reverse to 4×
[shift] + turn **[speed]**	knob is muted until release, then jumps straight to the new speed
[link] up	right deck mirrors left · saves and loads stereo
[shift]	adds a playhead at start (up to 4)
◆ multi-tap presets	

02 recording — lbh_02

[erase] + **[rec]**	**punch-in**: replaces audio while recording
[shift] **[shift]** + **[rec]**	**input monitoring**: on (rec lit) › armed (rec pulses) › off
armed **[rec]**	arm › record › stop & play · **[erase]** disarms, loop kept
[shift] + opp. **[rec]**	looping ← → **one-shot record** (records one pass, then stops)
[rec] jack	trig starts / stops recording, like **[rec]** (latching)
[rec] jack	gated: records only while the gate is high
◆ rec jack = gated	

* overdub keeps the old layer at 0.9 by default, so layers fade a little each pass (dub level, 04).

__fig.01 : speed

white speed LEDs = notches



4× rev	½ rev	stall	½	1×	2×	4×
+24 st	−12 st	–	−12 st	0 st	+12 st	+24 st

speed sets pitch: every doubling = +1 octave · reverse plays backwards

03 playback — lbh_03

[shift] + opp. **[erase]**	looping ← → **one-shot playback** (rec blinks white / mixed)
[shift] or **[retrig]** jack	one-shot: play one pass of start › length
touch the reel	warble or stall the tape · trimmer sets sensitivity
[speed] cv	±5 v, summed with the knob
[erase] + opp. **[erase]**	flip playback direction · speed LEDs blink
◆ v/oct speed mode only	
[speed] cv	tracks 1v/oct: sequence the loop's pitch
◆ v/oct · seq presets	

04 time knob — lbh_04

[shift] + **[rec]**	cycle the time function (amber LED by **[clk out]** shows which)
LED off	**clock divisions**: triggers per loop at **[clk out]**: 1–12, 16, 24, 32, 64
LED solid	**dub level**: how much old audio survives each overdub. high = delay feedback
LED pulsing	**user parameter**: crossfade, tape or slew (preset picks)



05 start · length — lbh_05

turn **[start]**	where the loop begins inside the recording
turn **[length]**	how much plays before jumping back. tiny = wavetable-like tones
[shift] + turn **[time]**	**quantise** start & length: lit LED sections = 1/2, 1/4, 1/8 …
[shift] + turn **[start]**	fine move that ignores the quantise grid (same for **[length]**)



ticks = quantise grid (1/8)

06 save · load — lbh_06

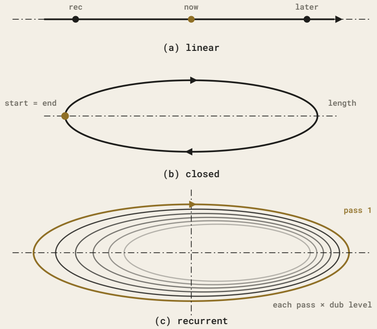
[shift] **[erase]** + **[rec]**	**save** to usb › saved (folder 0, newest first)
[shift] + **[erase]**	**load menu**
in the menu	own **[time]** = loop · opposite **[time]** = folder · **[rec]** loads · hold **[shift]** previews 5 s · **[erase]** exits
[link] up	saves and loads stereo (mono loads of stereo are summed)
power-on	last session reloads automatically

07 presets — lbh_07 · v2.1

[shift] **[shift]** + **[erase]**	**preset menu** for that deck
turn **[time]**	pick 1 of 12 (shown on the LED ring) · **[rec]** loads · **[erase]** cancels
usb › presets	.txt files, sorted 01–12. build your own with the preset editor

__fig.02 : the return

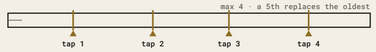
mm_02



08 multi-tap — lbh_08

◆ multi-tap presets only

[shift]	new playhead from start
[erase] + **[shift]**	delete the taps
[erase] trig, cv exp.	removes taps remotely



09 sync · clock — lbh_09

[clk out]	triggers per loop (set by clock div)
[retrig] jack	resets the playhead to start: sync to an external clock
[rec] jack	external clock sets the loop length
◆ clockable preset	
[erase] + **[rec]**	tap tempo instead of punch-in
◆ erase + rec = tap tempo	

10 aux · routing — lbh_10

aux in fader	sends **[aux]** in to the left ← → right deck input (+ **[cv]** ±5 v)
aux out fader	blends left ← → right outputs at **[aux]** out (+ **[cv]** ±5 v)
rear switch	aux out fader pre or post the output level, per deck



11 reset · calibrate — lbh_11

[shift] **[shift]** + **[rec]** **[rec]**	**reset modes** to default · audio kept · speed LEDs pulse
power on, hold **[shift]**×2 **[erase]**×2	**calibration**: speed knobs min, 0 v › **[rec]** each deck, then 1 v and 3 v; knobs max, no cv › **[rec]** = 5 v
amber blinks	voltage out of range: defaults used

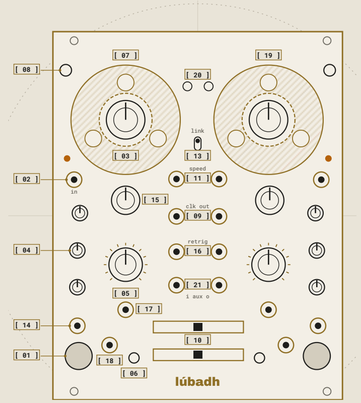
__fig.03 : front panel

01–11 = sections · 13–21 = hardware

left deck labelled · right deck mirrors it

[01]	**__rec :** record · overdub · stop
[02]	**__in :** input + level
[03]	**__speed :** rate & direction
[04]	**__time :** clock div · dub · user
[05]	**__start :** loop start
[06]	**__erase :** erase · save / load
[07]	**__LED ring :** position · presets
[08]	**__retrig/shift :** restart · shift
[09]	**__clk out :** clock triggers
[10]	**__aux faders :** in / out crossfade
[11]	**__speed cv :** ±5 v · 1v/oct

[13]	**__link :** stereo mirror
[14]	**__out :** output + level
[15]	**__length :** loop length
[16]	**__retrig :** jack, restart
[17]	**__cv :** aux fader cv
[18]	**__rec :** jack, record gate
[19]	**__reel :** capacitive touch
[20]	**__speed LEDs :** notches
[21]	**__aux :** in · out jacks



_where things hide

12 · lbh_12

preset files

usb › presets › .txt · instruo preset editor

- **speed** record variable · fixed, play follow · hold
- **speed control** notched · stepped · smooth · v/oct, markers 0–4
- **multi-tap** engine off · on
- **time user** crossfade · speed slew · tape emulation
- **rec jack** latching · gated · external clock
- **ext clock** average, resolution, timeout, divisions
- **quantise** all · even · odd · powers of two
- **erase + rec** punch-in · tap tempo
- **timing** min loop, crossfade, retrig delay, max dub
- **tape** wow / flutter, crinkle, age, hysteresis, wear, saturation, reverb
- **input filter** low cut freq + q, high cut
- **touch** off · stall · dip

usb disk

- **samples** 12 folders × 12 loops
- **saved** folder 0, newest first
- **presets** first 12 show in the menu

tails & tags

250 ms, in the file name

- {s} silent tail (default): keeps timing
- {n} no tail: smoothest, timing loosens
- {c} copy tail: keeps timing, may click

hardware

- **cv expander** **[length]** **[start]** **[time]** cv + **[erase]** trig, per deck
- **rear pcb** touch on / off · pre / post output level
- **front** touch sensitivity trimmer

factory presets

- **01–05** tape looper · clean · classic · multi-tape · clockable
- **06–10** octave delay · seq mono · seq poly · melontron · broken tape

tails at the loop point

250 ms · gold = the crossfade

